hoBIT: A Profile-Aware Retrieval-Augmented Chatbot for University Academic Advising

Anonymous ACL submission

Abstract

In university academic advising, identical questions have different correct answers depending on a student's department, admission cohort, and degree program. We present **hoBIT**, a profile-aware RAG chatbot that replaces a deployed rule-based FAQ predecessor at a university college of informatics. It treats a schema-typed per-session profile as a first-class retrieval signal: *offline*, it tags the corpus with profile metadata; *online*, it acquires the profile from the student by *dual-asking* and injects it into retrieval as a *soft prefix* and a *hard filter*. On a 3,400-case, three-track evaluation we report three findings: (i) once the profile is present, *profile-blind* reranking and HyDE become redundant; (ii) the gain is a hard-filter / soft-prefix synergy, the soft prefix dominating; and (iii) the pipeline holds on open-weight models—more faithful on a judge-free metric—for low-cost on-premise deployment. The benchmark is intentionally profile-dependent; our contribution is not showing that personalization helps, but demonstrating how to realize its benefit in practice and which profile-blind techniques become unnecessary.

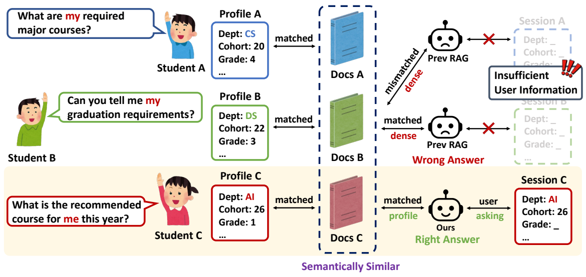

Figure 1: Profile-dependent advising. Students from different departments and cohorts (CS/20, DS/22, AI/26) ask first-person questions whose answers lie in *semantically similar* curriculum documents. Unlike a profile-blind retriever (*Prev RAG*), hoBIT (*Ours*) acquires the profile via *dual-asking* and retrieves the correct document.

1 Introduction

Retrieval-augmented generation (RAG) is increasingly tailored to specific domains (Gao et al., 2023b)—from medical education (Jang et al., 2025) to university student support (Trienes et al., 2025). Academic advising is one such domain: the *correct* answer to a recurring question—graduation requirements, required courses, double-major rules—is fixed not by any single attribute but by the *combination* of a student's department, admission cohort, and major type, so two students asking the identical question routinely need different curriculum tables (Figure 1).[1] The space is large: several majors—including convergence tracks such as brain-and-cognitive-science and information-security—each on a yearly-revised curriculum give dozens of valid department × cohort × major-type configurations that keep growing, beyond what hand-written rules can cover or the year-old FAQ chatbot can tailor to.

This is a *profile-dependent* retrieval problem: the corpus already contains the right chunk (Lewis et al., 2020), so the bottleneck is not recall but *disambiguation by user-side structure*. Because curriculum documents are *semantically near-identical* across cohorts, a profile-blind dense retriever surfaces the wrong one—exactly where the rule- and FAQ-based RASA (Bocklisch et al., 2017) predecessor also fails on first-person queries.

The disambiguating signal is a compact, schema-typed profile whose *few fields nonetheless index this large space*; hoBIT treats it as a *first-class retrieval signal*. Prior work personalizes differently—free-form personas from history (Zerhoudi and Granitzer, 2024), user-specific preference learning (Choi et al., 2025), or shared admin-curated content in university assistants (Trienes et al., 2025)—but none conditions retrieval on an *explicit, schema-typed* profile acquired on demand, the regime advising requires.

[1] CS, DS, and AI denote the Computer Science, Data Science, and Artificial Intelligence departments, respectively.

hoBIT acquires the profile without friction: it is *session-based* rather than login-based (short, one-off queries make accounts a friction and privacy liability), filling a TTL-bounded profile on demand by *dual-asking*—*proactively* when the predicted intent flags a field, *reactively* when a cited chunk still needs one—and injecting it into hybrid dense–sparse retrieval as a *soft prefix* and a *hard filter*. Our claim is not that personalization helps—expected on a profile-dependent benchmark—but that profile injection *reshapes what else matters*: it renders profile-blind reranking and query reformulation redundant, decomposes into a filter–prefix synergy, and holds on open-weight models for low-cost on-premise deployment.

We organize the evaluation around four research questions:

RQ1 **Dominance.** How large is the profile contribution relative to the pipeline's other retrieval components?

RQ2 **Subsumption.** Does profile injection render second-stage reranking and query reformulation (HyDE) unnecessary?

RQ3 **Mechanism.** What drives the gain—hard payload filtering or soft query-prefix injection?

RQ4 **Deployability.** Does the pipeline hold with open-weight retrievers and generators, enabling low-cost, on-premise deployment?

Contributions. **(1)** hoBIT, a deployment-ready profile-aware advising chatbot that enriches its corpus with profile-based metadata offline and, online, acquires a per-session profile by *dual-asking* and injects it as a hard filter and a soft prefix. **(2)** A 3,400-case, three-track evaluation suite and ablation harness over open and proprietary models. **(3)** Evidence that profile injection subsumes profile-blind reranking and HyDE, emerges from complementary hard filtering and soft-prefix conditioning, and generalizes to open-weight models for low-cost on-premise deployment. A live demo and source code are available.[2]

2 The hoBIT System

hoBIT centers on a schema-typed *user profile*—`department`, `major_type`, `grade`, `admission_year`, and `student_status`—as a first-class retrieval signal (Figure 2). The system follows an offline–online design. Offline, the corpus is enriched with profile-based metadata. Online, hoBIT acquires a session profile via dual-asking and injects it into a hybrid retriever as a hard metadata filter and a soft query prefix, focusing retrieval on profile-relevant evidence. Incoming queries are first routed by intent, with only retrieval queries entering the profile-aware pipeline, while the remaining intents are handled directly. Frequently asked administrative questions are answered from an administrator-maintained FAQ store, avoiding unnecessary LLM generation. Retrieved evidence is then passed to the generator, which produces grounded responses with sentence-level source citations. Additional implementation details are deferred to Appendix A.

Offline: profile-based semantic enrichment. Every chunk is LLM-tagged (OpenAI, 2024a) with two kinds of profile label: *value tags*—a `department` tag and an `admission_year` range—that the hard filter uses to exclude other cohorts, and a `required_profile` list naming which profile fields the chunk's answer depends on, which drives reactive dual-asking. Content is indexed in Qdrant (Qdrant, 2024) in lifecycle-split static/dynamic collections.

Online: acquiring the profile (dual-asking). Once a query is routed as a *retrieval* intent, hoBIT predicts *which* profile fields it needs—its `required_profile` (e.g., `department` and `admission_year` for "my graduation requirements?")—and checks them against the session. Missing fields are requested by *dual-asking*, at two points: **proactively**, before retrieval, for fields the prediction flags; and **reactively**, when a *cited* chunk's `required_profile` names a field the session still lacks (e.g., `major_type`). In either case hoBIT withholds the answer and asks for the missing field; the response is produced on the next turn, once the student supplies it. Values are cached in a Redis (Redis, 2024) session—no account or password—and reused silently, so each field is asked at most once.

Online: injecting the profile (hard & soft). hoBIT then injects the acquired profile at the two points where retrieval can use it. As a **soft prefix**, the profile is prepended to the query ("my graduation requirements?" → "Dept: CS, Cohort: 20; . . . "), steering dense and lexical matching toward the student's cohort. As a **hard filter**, the same profile constrains the candidate set through the chunk payload labels, dropping other-cohort

[2] https://hobit-emnlp.github.io/#demo

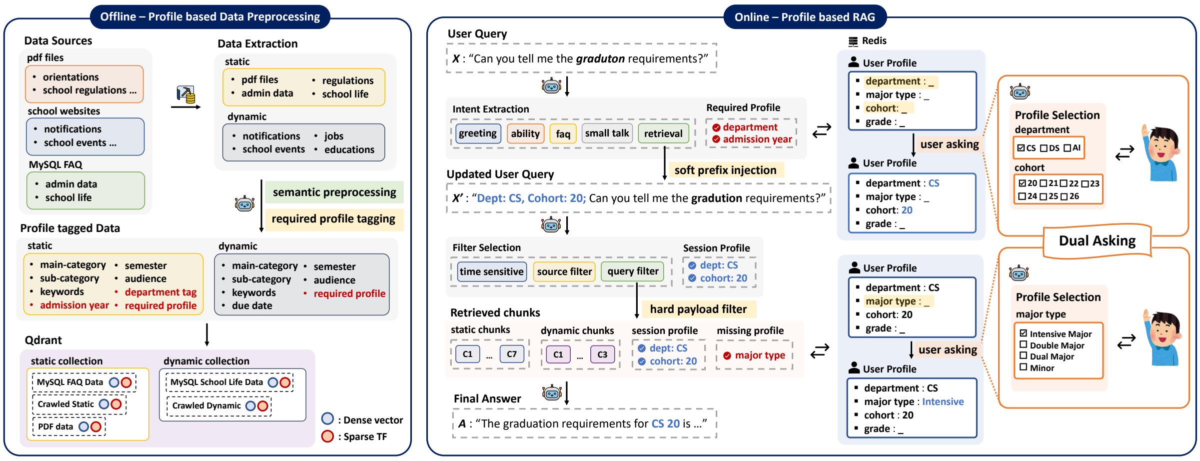

Figure 2: hoBIT centers on a schema-typed *user profile*. **Offline (left):** every chunk is LLM-tagged with the profile labels it presupposes—`department`, `admission_year`, and its `required_profile`—and indexed in Qdrant. **Online (right):** hoBIT predicts the profile fields a query needs, acquires missing ones by *dual-asking* into a Redis session, and injects the profile as a soft query prefix and a hard payload filter before generating a source-tracked answer; a still-missing field (e.g., major type) triggers a reactive ask.

chunks while keeping profile-agnostic content. The two are complementary—the soft prefix shapes *what ranks high*, the hard filter controls *what is eligible*. The generator then emits an answer with per-sentence `[Si]` citations.

		Retrieval		*Profile*	
Configuration	Component added	S/D	BM25	Filt.	Prof.
Dense	dense retriever (baseline)	–	–	–	–
+Lifecycle	static/dynamic index split	✓	–	–	–
+Hybrid	BM25 / RRF fusion	✓	✓	–	–
+Filtered	metadata payload filter	✓	✓	✓	–
hoBIT	profile injection (soft+hard)	✓	✓	✓	✓

Table 1: Ablation configurations (RQ1): each row adds one component—*S/D* static/dynamic split, *BM25*/RRF fusion, *Filt.* payload filter, *Prof.* profile injection. **hoBIT** is the full system; results in Table 2.

3 Experimental Setup

Datasets (three tracks, 3,400 unique queries). All tracks are built over a single college-crawled index, with categories curated from the corpus and validated against real usage logs (Appendix B). **(i) Intent routing** (1,600) classifies queries into five intents, with only *retrieval* queries entering the profile-aware pipeline. **(ii) Profile-grounded QA** (1,800), our main track, contains deterministic curriculum questions whose correct answers depend on the student's profile, enabling controlled evaluation of profile-aware retrieval. **(iii) Open-ended advising** (1,200) covers administrative and school-life queries without ground truth. Track (i) reuses the retrieval queries from (iii), giving 3,400 unique queries in total.

Ablation systems (RQ1). We compare five cumulative configurations of the same pipeline (Table 1), each introducing one additional retrieval component. This isolates the contribution of the static/dynamic split, hybrid retrieval, metadata filtering, and finally profile injection. The *Filtered* configuration enables the payload filter without profile information, serving as a control to measure the effect of filtering alone before profile injection.

Model pools (RQ2–RQ4). **RQ2** evaluates two profile-blind retrieval boosters: a cross-encoder reranker (BAAI, 2024) and HyDE query reformulation (Gao et al., 2023a). **RQ3** ablates the profile-injection strategy by enabling only the soft prefix, only the hard filter, or both. **RQ4** replaces one component at a time while holding the remainder of the pipeline fixed. We compare the open embedders `BGE-M3` (Chen et al., 2024) and `Qwen3-Embedding` (Qwen Team, 2025a) against `text-embedding-3-small` (OpenAI, 2024b), and the open generators `Qwen3-8B` (Qwen Team, 2025b) (multilingual) and `EXAONE-3.5-7.8B` (LG AI Research, 2024) (Korean-specialized) against `gpt-4o-mini` (OpenAI, 2024a). For generator comparison, retrieval is fixed to the deployed hybrid (`openai_small`+BM25); retriever comparison uses rank metrics only. The full retriever sweep is reported in Table 4.

System	CtxP	CtxR	Kw.	Src.M	KO-F	PASS
Dense	0.665	0.851	0.599	0.030	0.901	2.1%
+Lifecycle	0.710	0.889	0.605	0.024	0.899	1.9%
+Hybrid	**0.749**	0.880	0.587	0.068	**0.905**	6.2%
+Filtered	0.746	0.885	0.591	0.070	0.903	6.8%
hoBIT	0.718	**0.894**	**0.701**	**0.585**	0.888	**47.4%**

Auxiliary track	Score	PASS
Intent routing (5-class acc.)	0.960	–
retrieval-intent F1 (pipeline gate)	0.990	–
Open-ended advising (Overall)	**0.796**	59.1%

Table 2: **Main (top) and auxiliary (bottom) results.** *Top*: RQ1 deployed ablation (1,800 cases; configs Table 1)—the profile step drives nearly all the gain. *Bottom*: 5-class intent accuracy (with the pipeline-gating *retrieval*-intent F1) and open-ended *Overall* (LLM-judge, no ground truth). CtxP/CtxR context precision/recall, Kw. keyword, Src.M Source Match, KO-F Korean faithfulness, PASS/Overall pass at ≥ 0.5; best **bold**.

Metrics and evaluation modes. We evaluate in two modes. **(a)** RQ1 runs the deployed system (dual *static* + *dynamic* collections), scored by a graded *Source Match*, Keyword Match, and `deepeval` (Confident AI, 2024) LLM-judge metrics. **(b)** RQ2–RQ3 use a controlled single-static-collection setup with binary rank metrics (MRR, R@1/5/10), while RQ4 compares generators using the same quality metrics as (a). Although the quality metrics are shared, the evaluation protocols differ, so the reported scores are not directly comparable. Metric definitions, judge configuration, and the Source Match grading are provided in Appendix D.

4 Results

On the main Profile-grounded QA track, hoBIT far outperforms the best non-profile system (Table 2); two auxiliary tracks confirm the surrounding pipeline (Table 2), and we dissect *why* the profile drives the main result through RQ1–RQ4.
Two auxiliary tracks confirm the surrounding pipeline is sound (Table 2, bottom): intent routing reaches 96.0% accuracy—0.990 F1 on the *retrieval* intent that gates the profile pipeline—and the ground-truth-free open-ended track scores 0.796 Overall. The profile-dependent gains dissected in RQ1–RQ4 are the contribution.

4.1 RQ1: Profile Injection Dominates

Because the benchmark is intentionally profile-dependent, RQ1 measures the upper bound achievable with complete profile information. The remaining questions are how to approach this upper

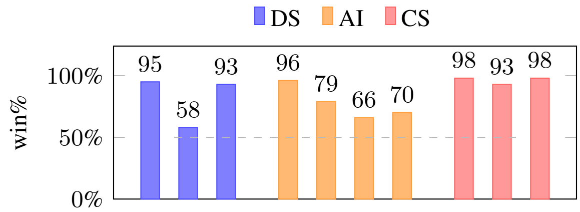

Figure 3: Blind pairwise study: per-pair win% of **hoBIT** over the Dense baseline, colored by department; dashed line at 50%.

bound in practice (§4.3) and whether profile injection makes profile-blind techniques such as reranking or HyDE unnecessary (§4.2). RQ1 assumes a complete profile, while Appendix C analyzes how closely the deployed dual-asking mechanism approaches this upper bound.

Profile injection dominates. Adding profile information (Filtered $\rightarrow$ hoBIT) is by far the most important factor, substantially improving both retrieval and answer quality (Table 2). In contrast, retrieval-only enhancements provide only marginal gains. Notably, enabling the payload filter without profile information has almost no effect, confirming that filtering is effective only when driven by user profiles. The improvements are most pronounced for cohort-specific questions, where the correct answer depends directly on the student's department, admission year, or major track.

Human preference. In a blind pairwise study (n=48, A/B counterbalanced, the *same* profile given to both systems), hoBIT beat the Dense baseline on *all* 10 profile-dependent pairs—an 85.3% aggregate non-tie win rate ($p < 0.001$), consistent across all three departments (Figure 3).

4.2 RQ2: Profile over Reranking and HyDE

We compare profile injection against two common RAG enhancements: reranking and query reformulation (HyDE) (Table 3).

Reranking and HyDE become redundant. A profile-blind reranker improves the no-profile baseline but degrades performance once profile information is available, as it can reorder the correctly retrieved cohort-specific curriculum by surface similarity. HyDE likewise cannot encode the student's department or cohort, providing little benefit without the profile and none after profile injection, while incurring an additional LLM call (~3 s/query). These results suggest that standard

Configuration	MRR	R@1	R@5	R@10	s/q
No profile					
baseline	0.038	0.009	0.069	0.152	–
+reranker	0.089	0.027	0.165	0.322	0.55
+HyDE	0.037	0.016	0.055	0.138	3.38
Profile					
+soft	0.276	0.151	0.464	0.614	–
+hard	0.189	0.112	0.280	0.433	–
hoBIT (soft+hard)	**0.538**	**0.462**	**0.652**	**0.711**	0.38
hoBIT+reranker	0.364	0.252	0.496	0.629	0.58
hoBIT+HyDE	0.431	0.326	0.569	0.683	3.31

Table 3: **RQ2–RQ3: controlled study** (single static collection; rank metrics over top-10). *No profile*: baseline plus profile-blind reranking/HyDE. *Profile*: the soft and hard channels, their combination **hoBIT** (soft+hard, RQ3), then the same boosters stacked on hoBIT. Best **bold**; *s/q* latency shown only where a stage adds real cost (– elsewhere; HyDE includes doc generation).

Retriever	MRR	R@1	R@5	R@10	R@50
BM25 (Kiwi)	0.425	0.318	0.594	0.706	0.926
`openai_small`	0.515	0.438	0.627	0.677	0.805
+BM25	**0.537**	**0.459**	**0.652**	**0.711**	**0.916**
BGE-M3	**0.605**	**0.552**	0.681	0.691	0.757
+BM25	0.593	0.525	**0.692**	**0.720**	**0.906**
Qwen3-Emb	**0.601**	**0.544**	0.685	0.704	0.784
+BM25	0.588	0.516	**0.693**	**0.731**	**0.906**

Generator	Src.M	Kw.	KO-F	Ovr.	tok	s/q
`gpt-4o-mini`	0.619	0.665	0.927	0.737	171	3.5
Qwen3-8B	0.653	**0.694**	**0.941**	**0.763**	252	7.1
EXAONE-7.8B	**0.667**	0.674	0.923	0.755	253	5.8

Table 4: **RQ4: open-model deployability** (1,800 cases). *Top*: retriever sweep—each embedder alone and as a shaded +BM25 (Kiwi (Lee, 2024)) hybrid, **bold** = larger of the two per column; `openai_small`+BM25 is deployed hoBIT. *Bottom*: generator sweep (shared retrieval), best per column **bold**. *tok* mean output tokens, *s/q* latency.

profile-blind reranking and query reformulation offer limited value once retrieval is conditioned on user profiles. We do not rule out profile-aware reranking; our finding is limited to conventional profile-blind methods.

4.3 RQ3: What Drives the Profile Gain

The profile is injected as a *hard* payload filter and a *soft* query prefix (§2); we decompose the two (Table 3).

Soft Prefix and Hard Filter Are Complementary. The soft prefix contributes more to recall than the hard filter by pulling the correct curriculum into the retrieved candidate set. The hard filter alone is less effective because the correct document is not always retrieved in the first place. However, achieving the best top-1 ranking requires both: the hard filter removes wrong-cohort candidates, while the soft prefix promotes the correct curriculum within the remaining set. This complementary behavior explains why the payload filter alone has almost no effect without profile information (§4.1).

4.4 RQ4: Open Models for Deployment

Open models are competitive. Open embedders match or outperform the proprietary baseline (Table 4, top), with BM25 hybridization consistently improving retrieval. Likewise, with retrieval fixed (Table 4, bottom), both open generators match or exceed `gpt-4o-mini`. Together, these results demonstrate that a fully open pipeline can be deployed without measurable loss in quality, although the open generators incur higher latency due to longer outputs.

Efficiency and deployment. Retrieval latency is dominated by embedding rather than search, so local open embedders run as fast as the remote API. While our deployed system uses `gpt-4o-mini` without exposing personally identifying information, the results indicate that a fully on-premise pipeline can be realized with no measurable loss in quality.

5 Demonstration

Figure 4 illustrates hoBIT's online interaction. We demonstrate two representative scenarios: a *cold start*, where the profile is acquired through *dual-asking*, and a *warm session*, where the cached profile is transparently reused.

UC1: Cold-start acquisition. A new student asks *"what are my required major courses?"* with no profile stored. hoBIT first acquires the student's `department` and `admission_year` through a *proactive* prompt, then requests `major_type` only when required by the retrieved curriculum (*reactive*). The completed profile is cached, and the system retrieves the correct curriculum with sentence-level source citations.

UC2: Warm-session reuse. The same student next asks *"what are my graduation requirements?"*. Because the profile is already cached, no further prompts are needed. The stored profile silently drives retrieval, returning the correct graduation requirements. The same query under a different profile retrieves a different curriculum.

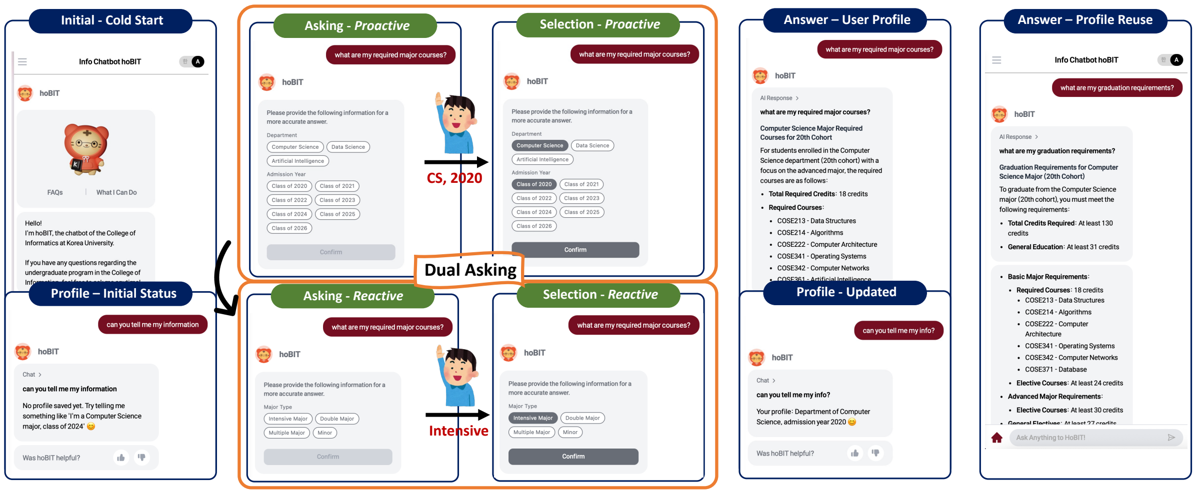

Figure 4: The hoBIT web interface: *dual-asking* acquires the profile at cold start (*proactive* then *reactive* ask-and-select), after which hoBIT answers with source citations and reuses the stored profile.

6 Related Work

Retrieval-augmented generation. RAG (Lewis et al., 2020) fuses dense (Karpukhin et al., 2020) and lexical (Robertson and Zaragoza, 2009) retrieval (Cormack et al., 2009), refined by reranking (Nogueira and Cho, 2019), query reformulation (Gao et al., 2023a), structured indices (Edge et al., 2024), or domain adaptation (Gao et al., 2023b; Jang et al., 2025)—all *profile-blind* boosters that §4.2 shows become redundant once a schema-typed profile conditions retrieval.

Personalization in LLMs. Personalization typically learns an *implicit* user model from *preferences* (Choi et al., 2025; Thonet et al., 2025; Yunusov et al., 2025), *history* (Qin et al., 2025; Li et al., 2025), or *personas* (Zerhoudi and Granitzer, 2024); hoBIT instead injects an *explicit, schema-typed* profile as a hard filter and soft prefix with no per-user learning, and Qin et al. (2025)'s value-over-similarity finding mirrors our RQ2 result that profile-blind reranking can *hurt*.

Academic-advising chatbots. Advising assistants have moved from rule-based dialogue (Bocklisch et al., 2017) toward retrieval-grounded LLMs (Jang et al., 2025). The closest, Marcel (Trienes et al., 2025), returns the *same* answer to every student; hoBIT is *profile-conditioned*, so identical questions yield cohort-correct curricula via dual-asking and source-tracked generation. To our knowledge it is the first *deployed* advising system to treat a schema-typed profile as a first-class retrieval signal and quantify it against reranking, hybrid, and query-side components (§4).

7 Conclusion

hoBIT is a deployed profile-aware RAG chatbot for academic advising. On a 3,400-case evaluation, profile injection *reshapes what else matters*: it renders profile-blind reranking and HyDE redundant, rests on a filter–prefix synergy, and holds on open-weight models for low-cost on-premise deployment. The profile itself is acquired friction-free by *dual-asking*—*proactively* before retrieval and *reactively* on a cited chunk—asking each field at most once.

Limitations

(1) Scope. The benchmark is built from a single college; although the schema and pipeline are domain-general, external validity remains to be established. **(2) Two evaluation modes.** RQ1 and RQ2–RQ4 use different evaluation protocols, so their scores are not directly comparable. **(3) Open-model deployment.** The open backend is validated offline but not yet deployed live; reported latencies are therefore indicative only. **(4) Profile correctness.** hoBIT relies on self-reported profile information without institutional verification.

Acknowledgments

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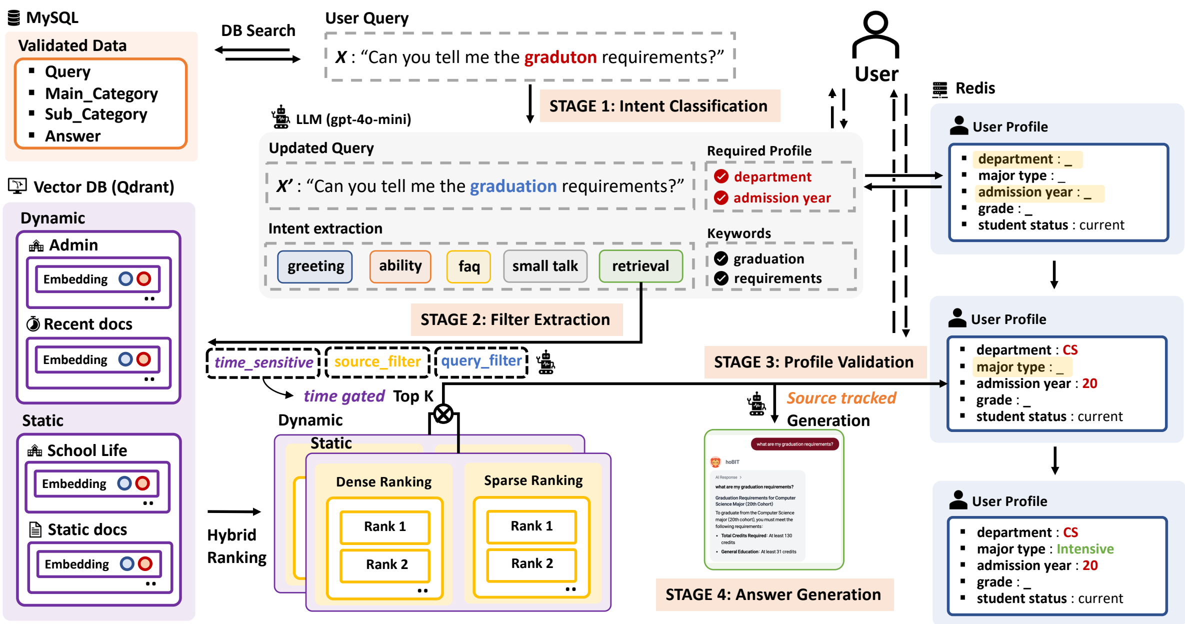

Figure 5: The four-stage runtime: **(1)** intent classification with a proactive profile check, **(2)** filter extraction and hybrid dense+sparse retrieval, **(3)** reactive validation on the cited chunks, and **(4)** source-tracked generation.

Appendix A Runtime Pipeline

The deployed system runs in four stages over two endpoints (`/classify` and `/chat`). While §2 describes the overall design, this section summarizes the implementation details (Figure 5).

Stage 1 (`/classify`). `gpt-4o-mini` predicts the intent, normalized query, keywords, and `required_profile`. For curriculum queries, `department` and `admission_year` are always required. If a required field is missing from both the query and session, the request is blocked before `/chat`, avoiding unnecessary retrieval and generation.

Stage 2 (`/chat`). The profile is injected through three channels: a payload filter over cohort tags (`department` and `admission_year`), a query prefix, and a system-prompt hint. Dense (`text-embedding-3-small`) and sparse (BM25/Kiwi) retrieval are fused by RRF (k=60), with results blended across the *static* (curriculum) and *dynamic* (notices) collections using a 70/30 weighting determined by query type.

Stages 3–4 (validation and generation). The generator attaches a `[Si]` citation to every sentence; otherwise it abstains. hoBIT then re-checks the `required_profile` of the cited chunks only. If additional profile information is required, the draft is discarded and a reactive prompt is issued. The same citations also power the UI source links and the Source Match metric (§Appendix D).

Appendix B Dataset

Index. A single index of 906 chunks from curriculum PDFs, orientation booklets, department webpages, and an operational MySQL FAQ, segmented and LLM-tagged as described in §2.

Real-usage anchors. From 3,058 historical RASA queries, filtering and retrieval validation produced 797 anchors. These are used only to validate the benchmark (Appendix B), never as generation prompts.

Curation. Categories were derived bottom-up from the index for reproducibility, removing sparsely supported cases; their real-usage coverage is checked below.

Track II (Profile-grounded QA, main track). The 1,800 cases fully cross 60 profiles × 10 profile-dependent categories (major-required courses, general education, graduation-credit breakdown, double-major rules, etc.) × 3 query styles (*formal*, *personal*, gold-keyword *verification*). The 60 profiles are 15 department–admission-year cohorts (CS, DS, AI) × 4 grades, with admission years bucketed into 8 *curriculum-revision* groups (e.g., CS in 2020/2021/2024/2025)—the grouping that fixes which curriculum is correct. Every case carries a deterministic gold source and expected keywords validated against the index (enabling Source

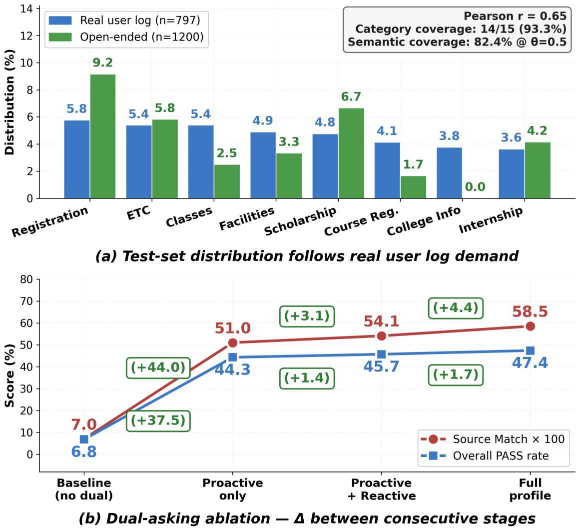

Figure 6: **(a)** Test-set category distribution vs. the real user log (three-axis alignment stats inset). **(b)** Dual-asking ablation—cumulative Source Match and PASS gains over the no-profile baseline: proactive dominates, reactive adds a marginal gain, a full (oracle) profile bounds the headroom.

Match, Keyword Match, and Recall@K), and the profile appears only in the session, never in the query text—so the ablation isolates profile-aware retrieval.

Tracks III and I. Track III contains 12 administrative and school-life categories with LLM-judged queries synthesized from reference chunks. Track I reuses its 1,200 retrieval queries and adds 400 non-retrieval queries, evenly split across greeting, ability, FAQ, and smalltalk.

Test-set realism. We validate the benchmark against the 797 in-scope log anchors on three complementary axes (Figure 6a). **Coverage:** 14/15 log categories (93.3%). **Proportionality:** top-8 category frequencies correlate with the log at Pearson r=0.65. **Semantic realism:** using the deployed embedding model (`text-embedding-3-small`), 82.4% of anchors have a benchmark query within cosine 0.5.

Appendix C Dual-Asking Ablation

Figure 6b decomposes dual-asking into its *proactive* (Stage 1) and *reactive* (Stage 3) components. **Proactive** acquisition of department and cohort before retrieval contributes the bulk of the gain—roughly 85% of the improvement over a no-dual baseline—since without these fields cohort-correct matching is essentially impossible. **Reactive** catch-up adds a smaller but consistent lift, fired on only the 24.3% of cases whose first response exposes a missing field (e.g., `major_type`), so its per-case effect is larger than the aggregate suggests. The remaining gap to the full-profile oracle reflects cases where the complete profile is still unavailable, motivating future improvements in profile acquisition.

Appendix D Metrics

Deterministic. *Keyword Match*: fraction of expected content keywords appearing in the response. *Source Match* (RQ1): graded attribution over the deployed dual collections—gold source cited / retrieved / missing = 1/0.5/0—rewarding citation rather than retrieval alone. *MRR* and *Recall@K* (RQ2–RQ3): standard ranking metrics over a single static collection, with Recall@K measured as binary hit@K.

LLM-judge (`gpt-4o-mini` via deepeval). *Contextual Precision/Recall*: relevance and coverage of the retrieved context. *Korean Faithfulness* (GEval): whether every claim is supported by the retrieved context (claim-free answers score 1.0). *Answer Completeness* and *Top-3 Precision*: answer coverage and top-3 retrieval quality. *Overall PASS*: all applicable metrics exceed 0.5.

Appendix E HyDE Example

For *"what is the credit difference between my basic and intensive major?"*, HyDE generates a fluent hypothetical document, but one that is profile-agnostic. Without the student's department and admission year, the rewritten query cannot distinguish between semantically similar curricula, retrieving a generic curriculum rather than the correct profile-specific one (§4.2).

GenAI Usage Disclosure

We use `gpt-4o-mini` as hoBIT's answer generator, as the LLM judge (via `deepeval`), and to generate the HyDE documents used in RQ2. Deterministic metrics are computed without LLM judgement. Prompt templates, evaluation settings, and the demonstration video are available on the project homepage. In addition, the authors used general-purpose AI assistants during paper drafting (organization and copy-editing) and as programming aids for portions of the implementation. All technical claims, system design decisions, experimental analyses, and final code review originate with and remain the responsibility of the authors.